Insurance Cost Analysis

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Dataset Overview

- Total Records: **1338 customers**
- Numerical Features: **age, bmi, children, charges**
- Categorical Features: **sex, smoker, region**
- Target Variable: **charges**
- Major cost-driving variable identified: **smoker**

Source and Integrity

- Cleaned dataset from **insurance.csv**, containing **1338 customer records**.
- Missing values were checked and handled to ensure data accuracy.
- Categorical variables (sex, smoker, region) were properly formatted.
- Dataset is reliable for statistical analysis and predictive modeling.

Key Fields for Analysis

- **age:** Customer age, used to analyze cost variation with age
- **sex:** Gender of the customer (male/female)
- **bmi:** Body Mass Index, important indicator of health risk
- **children:** Number of dependents covered by insurance
- **smoker:** Smoking status (smoker/non-smoker), major cost factor
- **region:** Residential region (northeast, northwest, southeast, southwest)
- **charges:** Medical insurance cost charged to the customer (target variable)

Scope and Purpose

- Customer Risk Analysis

Analyze how customer characteristics like age, BMI, and smoking affect insurance charges.

- Cost Distribution Insights

Identify how insurance charges vary across different customer groups.

- Risk Factor Identification

Find key factors responsible for high insurance cost using statistical methods.

- Predictive Modeling Support

Provide a foundation to build models that predict future insurance charges.

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Managing Increasing Insurance Expenses

The Problem : Variation in Insurance Charges

- Insurance companies face **significant variation in medical charges across different customer groups.**
- Insurance costs differ greatly based on key factors such as **smoking status, age, BMI, and number of children**.
- **Smokers and customers with higher BMI and older age show much higher average charges**, increasing overall insurance risk.
- A small group of customers has **extremely high charges (outliers), which increases financial pressure on insurance companies**.

Business Impact : Financial and Risk Consequences

- Customers with **higher medical charges significantly increase the total cost burden** on insurance companies.
- **Smokers, older individuals, and high-BMI customers contribute disproportionately to total insurance payouts**, making them high-risk segments.
- Without proper risk identification, insurance companies may **underprice or overprice policies**, leading to financial losses or reduced competitiveness.
- **Predictive modeling and data analysis are essential to identify high-cost customers early**, allowing better pricing, risk management, and cost control.

Major Insights from Data Analysis

Smoking as the Primary Cost Driver

- Smokers have significantly higher insurance charges than non-smokers.
- Average smoker charges are approximately **3-4 times higher** than non-smokers.
- Smoking is the **most influential factor affecting insurance costs** compared to age, BMI, and number of children.

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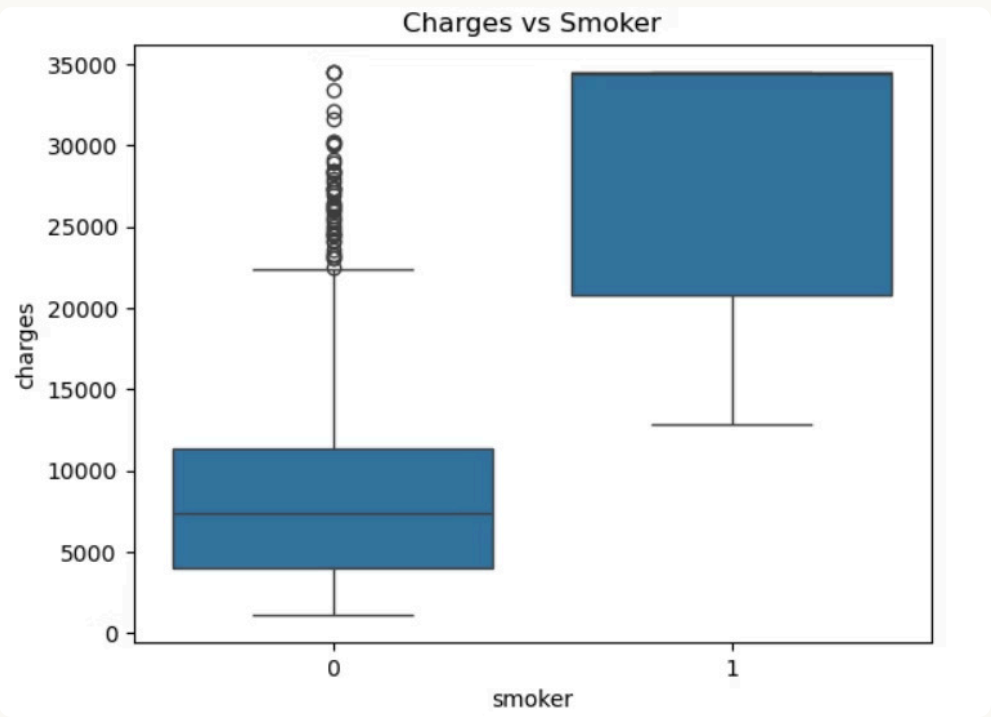
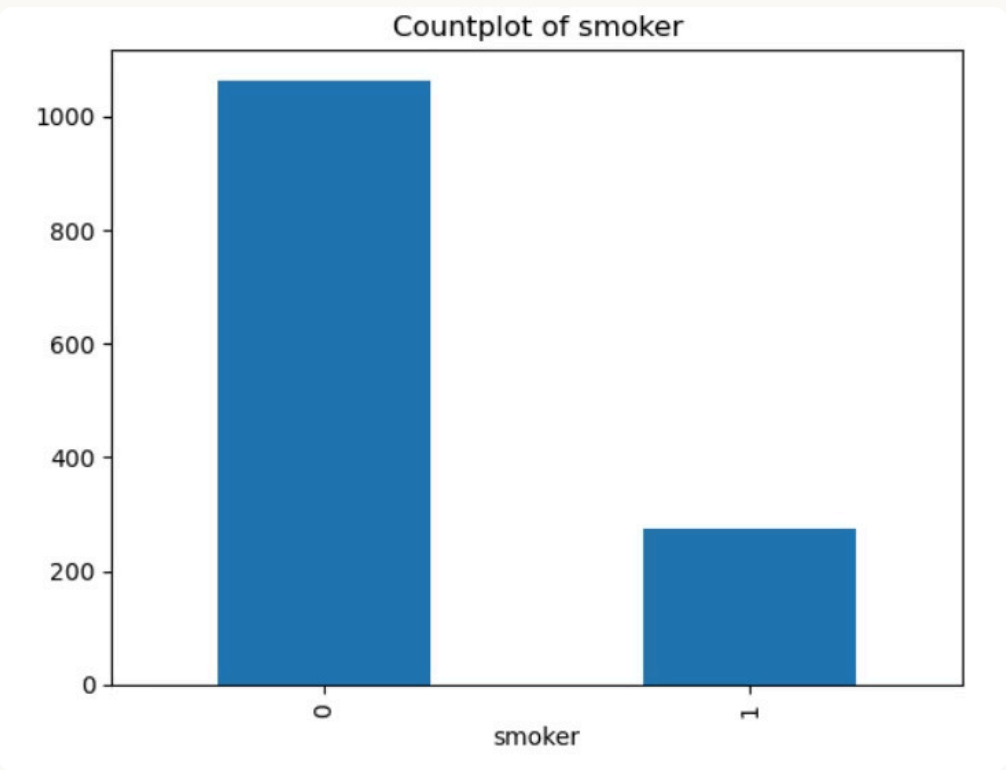
Insurance Cost Distribution: Understanding Cost Patterns

By Smoking Status

Insights:

- Smokers represent a smaller portion of customers but contribute disproportionately to total insurance costs.
- Average insurance charges for smokers are significantly higher than non-smokers.
- Smoking status is the strongest predictor of insurance charges.





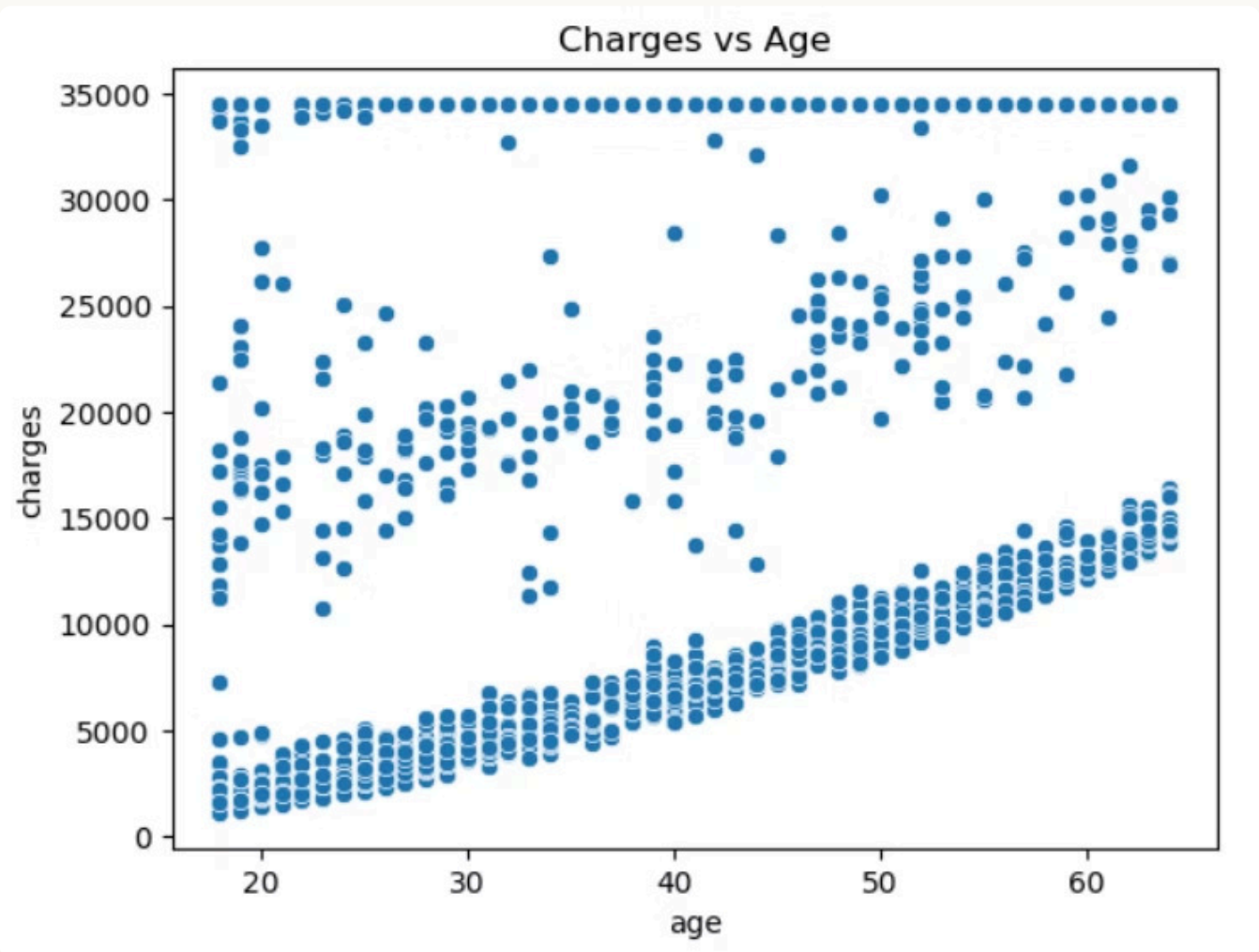
Graph Interpretation :

- Non-smokers are more in number, but smokers have much higher charges.
- Boxplot shows smokers have higher median and wider spread of charges.

By Age Group

Insights:

- Insurance charges increase steadily with age.
- Customers above age 50 show the highest average charges.
- Older customers represent higher medical risk and higher insurance costs.



Graph Interpretation:

- Scatterplot shows positive relationship between age and charges.
- Higher density of expensive claims among older individuals.

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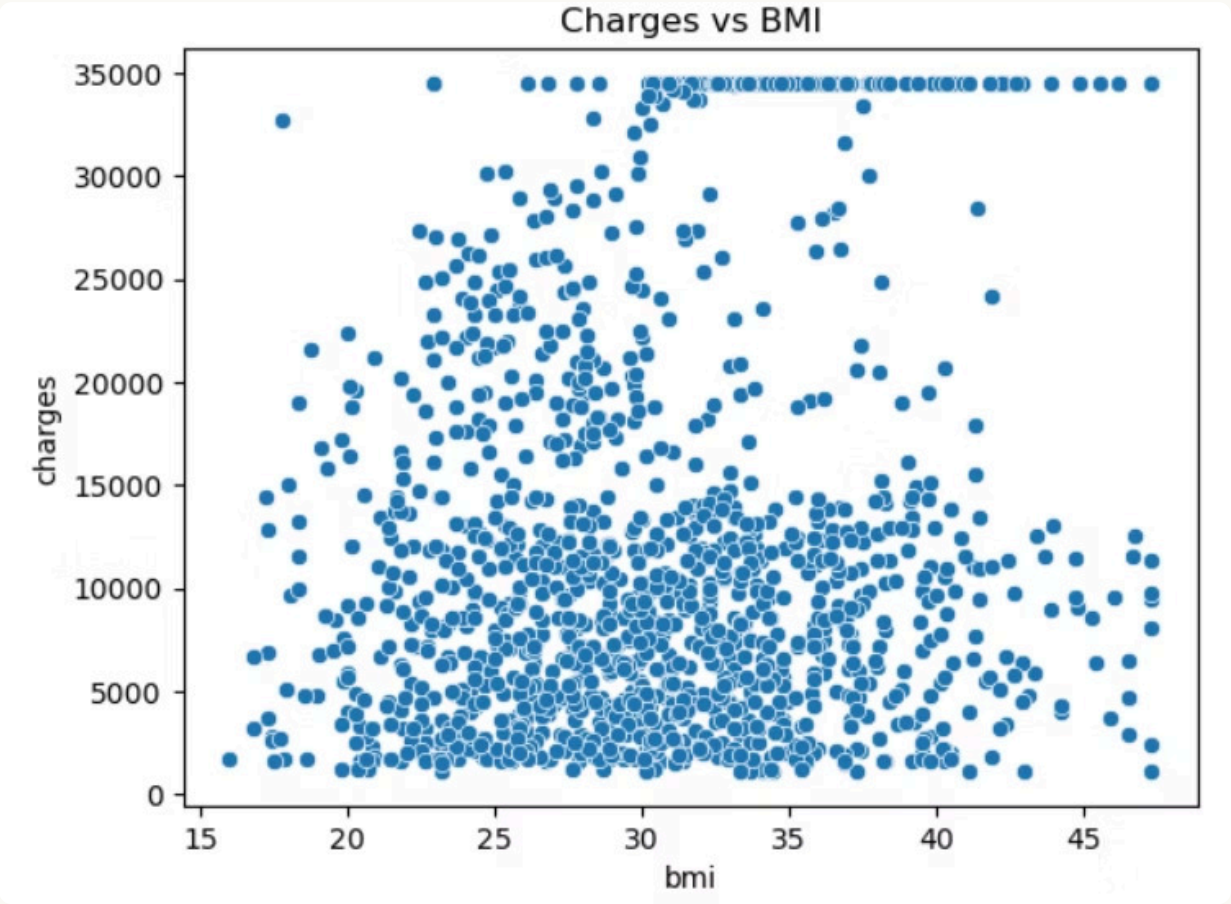
By BMI Category

Insights:

- Customers with higher BMI have higher insurance charges.
- Obese customers represent a high-risk and high-cost segment.
- BMI has a moderate positive impact on insurance costs.



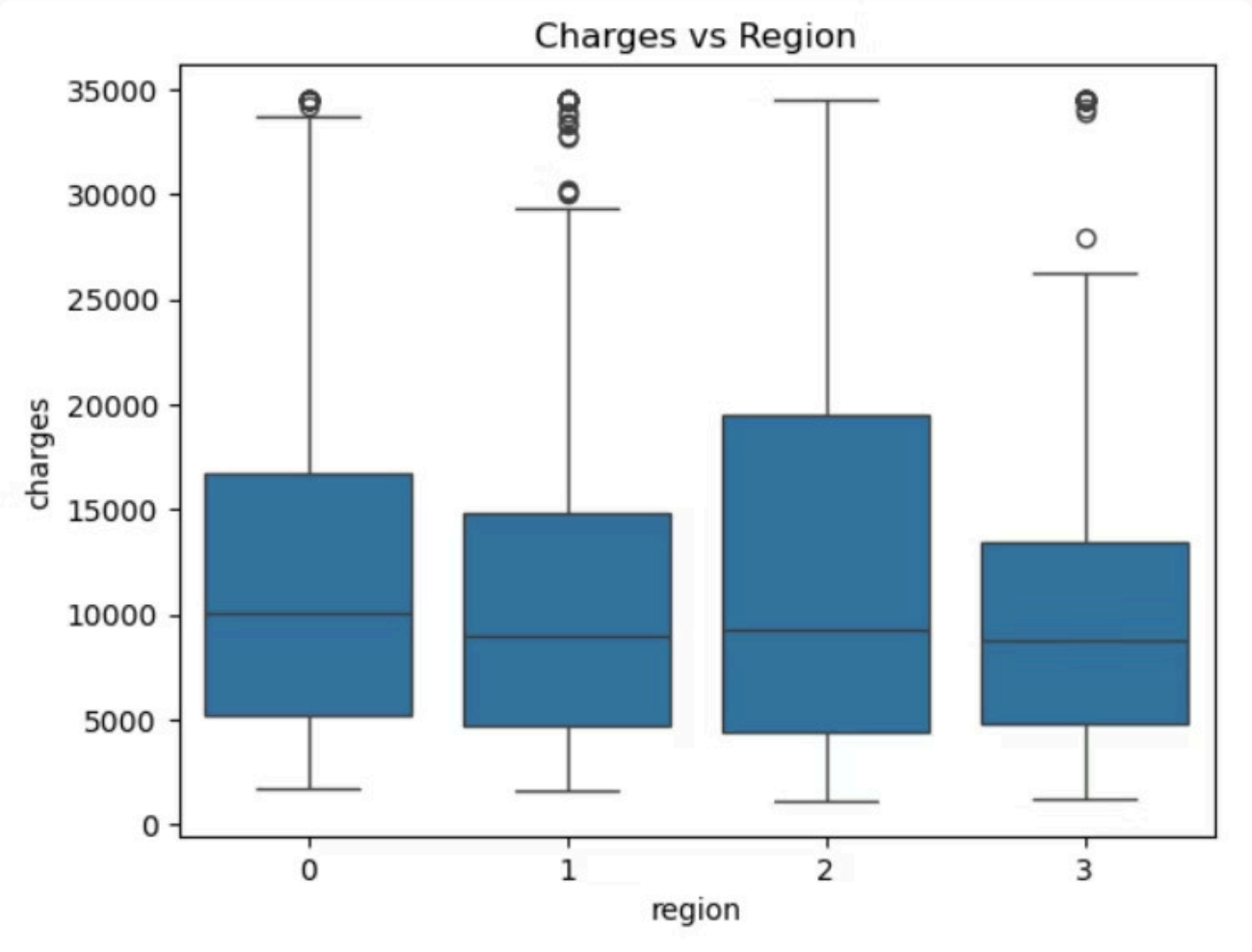
Graph Interpretation:

- Charges increase as BMI increases.
- High BMI values show more high-cost outliers.

By Region

Insights:

- Insurance charges are fairly evenly distributed across regions.
- Slight variations exist, but region is not a major cost driver.
- Smoking and BMI influence charges more than geographic location.



Graph Interpretation:

- Median charges are similar across regions.
- No strong regional effect on insurance cost.

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Key Factors Driving Insurance Costs

Smoking status

- Smokers pay much higher charges than non-smokers.
- Because smokers have higher health risks.

Age

- Older people have higher insurance costs.
- Because health risks increase with age.

BMI (Body Mass Index)

- Higher BMI (overweight/obese) leads to higher charges.
- Because obesity increases medical problems.

Number of children (dependents)

- More dependents can slightly increase insurance cost.

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Statistical Contribution Analysis

T-Test Result (Smoker vs Non-Smoker)

Smoking has a statistically significant impact on insurance charges. Smokers pay much higher insurance costs than non-smokers. The very high T-statistic (46.46) and extremely low p-value (0.0000) confirm that smoking is the strongest driver of insurance expenses.

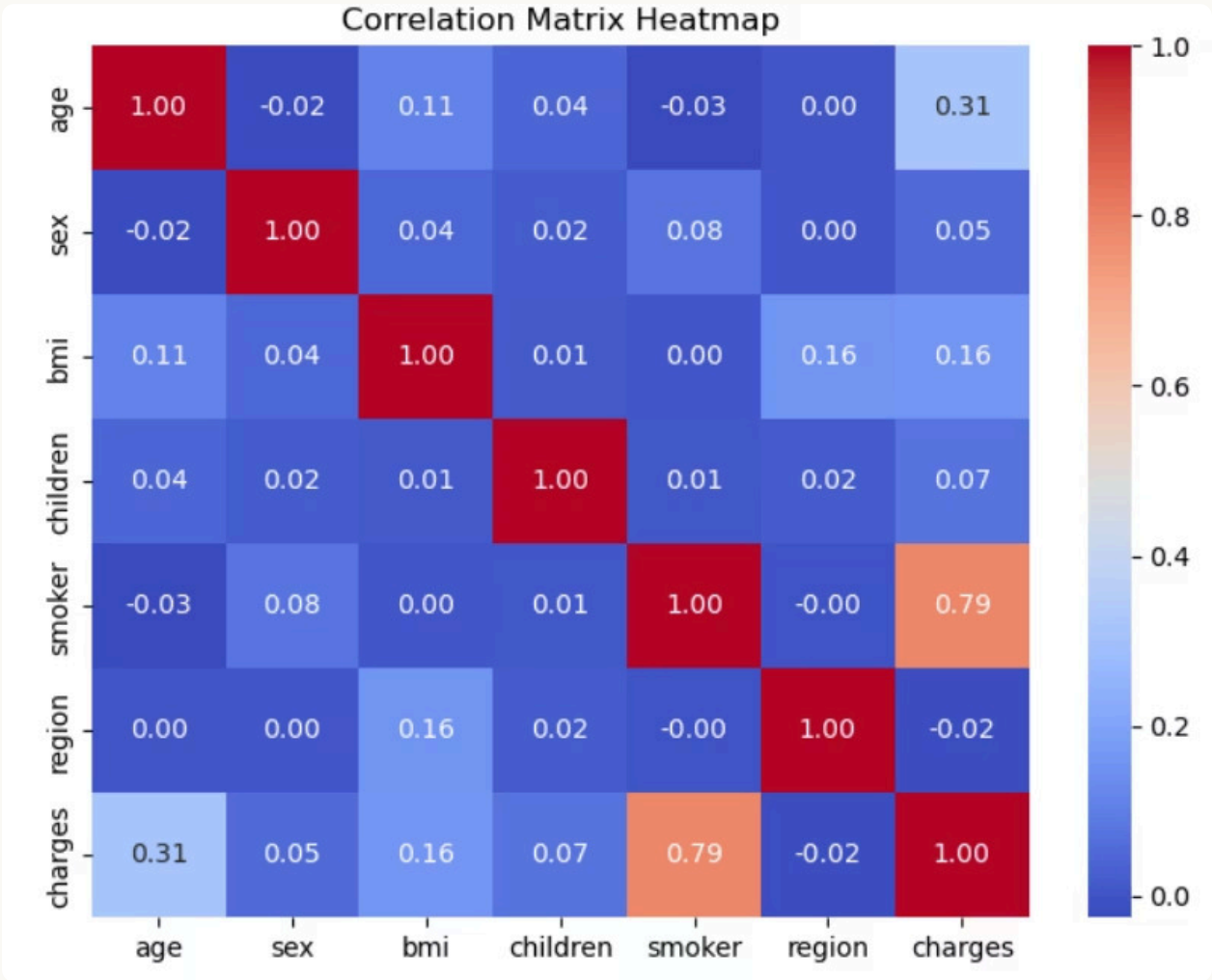
ANOVA Test Result (Region vs Charges)

ANOVA test result (F = 2.31, p = 0.0742) shows that insurance charges do not significantly differ across regions. This indicates that region is not a major factor influencing insurance costs compared to other variables like smoking, age, or BMI.

Correlation Analysis

Key Findings

- Smoking has the strongest positive correlation with insurance charges (0.79).
- Age has a moderate positive correlation (0.31).
- BMI has a weak positive correlation (0.16).
- Children, sex, and region have very weak or no correlation with charges.



Smoking shows the highest correlation with insurance charges, followed by age and BMI. Region, sex, and number of children have minimal impact on insurance cost variation.

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Financial Impact on Insurance Companies

- Smoking significantly increases medical costs, leading to higher insurance payouts.
- High-risk customers contribute disproportionately to total insurance expenses.
- Older individuals and customers with higher BMI also increase claim probability.
- Identifying these drivers helps insurers control overall financial risk.

Risk Segmentation Strategy

- Smokers represent the highest risk customer group.
- Older customers (age 50+) show consistently higher insurance charges.
- Customers with high BMI have elevated medical risk and costs.
- These groups should be categorized as high-risk segments.

Pricing and Policy Optimization

- Insurance companies should adjust premiums based on customer risk profile.
- Risk-based pricing ensures financial sustainability and fair premium distribution.
- Early identification of high-risk customers improves profitability.
- Predictive modeling can automate risk classification.

Strategic Decision Support

- Smoking is the strongest predictor of insurance cost (confirmed by T-test and correlation).
- Region has minimal impact, so geographic pricing adjustments are less necessary.
- Data-driven pricing improves competitive advantage and risk control.

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Predictive Modeling Approach for Insurance Cost

Linear Regression

- Predict insurance charges using age, BMI, smoking status, etc.
- Simple and easy to interpret.
- Helps understand how each factor affects cost.

Ridge and Lasso Regression

- Improve prediction accuracy by reducing overfitting.
- Automatically select important features.
- Useful when multiple variables influence charges.

Tree-Based Models (Decision Tree / Random Forest)

- Capture complex and non-linear relationships.
- Identify high-risk customer segments.
- Provide higher accuracy compared to linear models.

Business Value

- Accurate premium prediction
- Better risk management
- Increased company profitability

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Conclusion and Final Recommendations

Key Findings Summary

- Smoking is the most significant factor affecting insurance charges.
- Smokers pay significantly higher insurance costs than non-smokers (confirmed by T-test and correlation).
- Age and BMI also have a moderate impact on insurance expenses.
- Region has minimal influence on insurance charges (confirmed by ANOVA test).

Use Predictive Modeling

- Implement machine learning models to predict insurance charges.
- Automate risk identification and pricing decisions.

Data-driven analysis helps insurance companies accurately predict costs, identify high-risk customers, and optimize pricing strategies for sustainable business growth.

Thank You.

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